

# Rachel Kan

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## Education

**Cornell University, College of Engineering, GPA: 3.87**

Ithaca, NY

Bachelor of Science in Computer Science • Intended Minors: Artificial Intelligence, Business

Expected May 2028

**Related Courses:** Honors Object-Oriented Design and Data Structures, Data Structures and Functional Programming, Discrete Structures, Probability and Statistics, Foundations of AI Reasoning and Decision-Making, Intro to Machine Learning, Intro to Analysis of Algorithms, Language and Information, Computer System Organization and Programming

## Technical Skills

- **Programming:** Python, Java, OCaml, C, JavaScript, HTML/CSS, QML, JSON, RESTful APIs
- **Frameworks/Tools:** Git, AI-assisted development, Jupyter Notebook, VS Code, PyTorch, LaTeX, IntelliJ, React

## Project Experience

**Knowledge Graphs for Solar Energy News Analysis**

June-August 2026

Research Assistant, Syracuse University Text Analytics for Social Scientists (NSF/DOD-funded)

Syracuse, NY

- Engineered a Python pipeline that transformed unstructured solar energy news into a 4,002-node, 5,127-edge knowledge graph using GPT-4o, while reducing LLM inference costs through domain-specific filtering
- Automated preprocessing of the Cornell Newsroom dataset with Pandas, NLTK, and NumPy, reducing LLM inference costs through domain-specific keyword filtering and data transformation
- Implemented entity resolution using Sentence-Transformers semantic embeddings and RapidFuzz similarity matching before constructing graph representations with NetworkX
- Applied Clauset-Newman-Moore community detection with Networkx to identify 408 communities and analyze connectivity across the global solar energy ecosystem and present findings at a university research poster session

**AI Sector Financial Analysis and Sentiment Modeling for AI Bubble prediction**

August 2025-May 2026

Quantitative Student Analyst, Millennium x WICC Project

Ithaca, NY

- Collaborated in a 4-person team under the guidance of a Millennium Technology Project Lead to conduct analysis of trends in AI sector stocks and company valuations and presented findings at Millennium headquarters
- Applied RoBERTa and VADER natural language processing frameworks to public data streams (e.g. Google News, Yahoo Finance, and social media) to correlate shifts in public sentiment with AI market fluctuations
- Evaluated AI bubble risks by using Python (yfinance) to index stock trajectories against the Dot-Com crash; synthesized capex, revenue multiples, and cash flows for market leaders (NVIDIA, Microsoft, OpenAI, etc.)

**Real-Time Command and Control Graphical User Interface (GUI)**

June 2025-August 2025

Software Engineering Intern, Advanced Technology Systems Company (ATSC-IES)

Gilbert, AZ

- Designed and built a touchscreen/desktop Qt-based GUI integrating real-time map views and RTSP camera feed
- Visualized MIL-STD-2525 detection data with dynamic updates, clustering, and contextual interaction
- Built spatial analysis tools (e.g., pin drop, viewshed, tape measure) with intuitive manual and drag-edit support
- Developed C++ RESTful APIs to enable spatial analysis tools and real-time PTZ camera control (JSON-over-RTSP) for pan, tilt, zoom, and visual settings
- Leveraged AI-assisted coding tools for rapid development, debugging, and continuous code quality improvements

## Extracurricular Activities

**Women in Computing at Cornell (WICC) Executive Board Member**

January 2025-Present

- Underclassmen Outreach Co-Director

August 2025-May 2026

- Vice President: Academic Team

May 2026-Present

**CS 4820/5820: Introduction to Analysis of Algorithms Teaching Assistant**

January-May 2026

**CS 3700/5700: Foundations of AI Reasoning and Decision-Making Teaching Assistant**

Fall 2026

## Skills and Interests

**Language Skills:** Conversational Fluency in Mandarin, Professional Fluency in Spanish

**Interests:** Nonograms, Nail art, JoJo's Bizarre Adventure